

4.1 Equations of Motion and the Dielectric Function

- Assume that: An ensemble of identical uncoupled harmonic oscillators, neglecting damping for the moment.

- Then eigen-frequency ω_0' :
 $\omega_0'^2 = \beta m^{-1}$ (4.1)

- Fig.4.1b: $\lambda = \infty$ or $k = 0$ (4.2a)

- Fig.4.1c: $\lambda_{\min} = 2a$ or $k_{\max} = \frac{\pi}{a}$ (4.2b)

1. I-point: $k = 0$ gives the center of the first Brillouin zone

2. (4.2b): the shortest physically meaningful wavelength and the largest k -vector gives the boundary of the first Brillouin zone for a simple linear chain or cubic lattice.

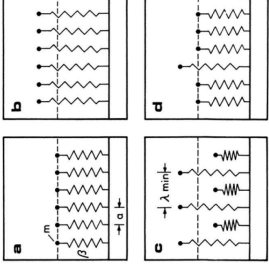


Fig. 4.1. A part of a periodic array of uncoupled oscillators in their equilibrium position (a); elongated with a wavelength $\lambda = \infty$ (b); with the shortest physically meaningful wavelength $\lambda_{\min}$ (c); and a non-propagating wave packet (d). After [6dH]



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Uncoupled and coupled oscillators

zero coupling $\leftrightarrow$ zero bandwidth $\leftrightarrow v_g = 0$, Fig.4.2a

finite coupling $\leftrightarrow$ finite bandwidth $\leftrightarrow v_g \neq 0$, Fig.4.2b

(4.3)

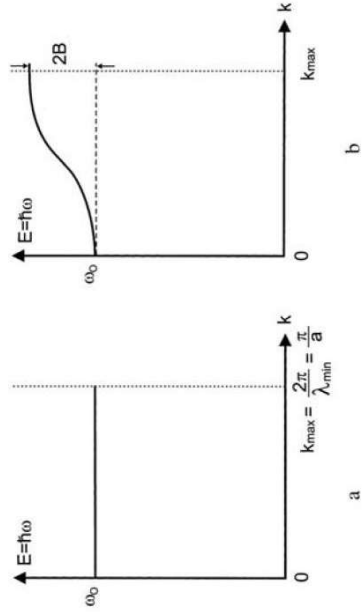


Fig. 4.2. The dispersion relation of an ensemble of uncoupled (a) and of coupled (b) oscillators



CHAPTER 4

Ensemble of Uncoupled Oscillators

- The optical properties of semiconductors are determined by the interaction of various types of oscillators and the electromagnetic field
- An incident electromagnetic field will cause oscillators to perform forced oscillations – depend on
 - the frequency of the incident light
 - the eigenfrequency of the oscillators
 - the coupling strength
 - the damping of the oscillation



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Main content

- 4.1 Equations of Motion and the Dielectric Function Based on Classic Theory
- 4.2 Corrections Due to Quantum Mechanics and Local Fields
- 4.3 Spectra of the Dielectric Function and of the Complex Index of Refraction
- 4.4 The Spectra of Reflection and Transmission



4.2 Corrections Due to Quantum Mechanics and Local Fields

- In classic mechanics model, the term $N\epsilon^2 m^{-1}$ gives the coupling strength
- In quantum mechanics, this coupling is given by the square of transition matrix element for dipole-allowed transitions:

$$|H_{ij}^{(p)}|^2 = |\langle j | H^{(p)} | i \rangle|^2 \quad (4.15)$$

- Usually define the oscillator strength: $f = \frac{2N\omega_0'}{\epsilon_0\hbar} |H_{ij}^{(p)}|^2$ and

$$\epsilon(\omega) = 1 + \frac{f}{\omega_0'^2 - \omega^2 - i\omega\gamma} \quad (4.16)$$

- For dilute systems, E is just the external incident field and use (4.17) as it is

- For dense systems, consists of two parts = external field and the field created by all the other ~~dipoles~~ $\epsilon(\omega)$ can be written:

$$\epsilon(\omega) = 1 + \frac{f}{\omega_0'^2 - \omega^2 - i\omega\gamma} \quad (4.17)$$

with a shifted eigenfrequency:

$$\omega_0'^2 = \omega_0'^2 - \frac{Ne^2}{3m\epsilon_0} = \omega_0'^2 - \frac{f}{3} \quad (4.18b)$$

介电函数有频率

4.3 Spectra of the Dielectric Function and of the Complex Index of Refraction

- A semiconductor contains not only one type of oscillators
- Helmholtz-Ketteler formula or Kramers-Heisenberg dielectric function:

$$\epsilon(\omega) = 1 + \sum_j \frac{f_j}{\omega^2 - \omega_{0j}^2 - i\omega\gamma_j} \quad (4.20a)$$

If the eigenfrequencies ω_0 form a continuous band:

$$\epsilon(\omega) = 1 + \int \frac{f(\omega_0)}{\omega^2 - \omega_0'^2 - i\omega\gamma(\omega_0)} d\omega_0 \quad (4.20b)$$

For simplicity, we use (4.20a) in the following.

- For a single resonance ω_{0j} well separated from all other resonances, the contribution of other resonance $\omega_{0,j'}$ in the vicinity of ω_{0j}

$$\text{For } \omega \ll \omega_{0j}, \quad \frac{f_j}{\omega_{0j}^2} = \text{const} \quad (4.21)$$

and for the contribution tends to zero for $\omega \gg \omega_{0j}$

So we can neglect the lower resonance and summarize all higher resonances in a so-called background dielectric constant ϵ_b

So we get the simplified expression in the surroundings of ω_{0j} :

$$\epsilon(\omega) = \epsilon_b + \frac{f_j}{\omega^2 - \omega_{0j}^2 - i\omega\gamma_j} \quad (4.22a)$$

- The electric field of incident light on the independent oscillators:

$$\mathbf{E} = (E_0, 0, 0) \exp[i(k_z z - \omega t)] \quad (4.4a)$$

- Considering the oscillator at $z = 0$:

$$\mathbf{E} = (E_0, 0, 0) e^{-i\omega t} \quad (4.4b)$$

- Assume every oscillator carries a charge e , then a force by $\mathbf{E}$:

$$\mathbf{F} = eE_0 e^{-i\omega t} \quad (4.5)$$

The equation of motion is:

$$m\ddot{x} + \gamma m\dot{x} + \beta x = eE_0 e^{-i\omega t} \quad (4.6)$$

The general solution of (4.6) = the general solution of the corresponding homogeneous equation + a special solution of the inhomogeneous one.

$$\text{Assume: } x(t) = x_0 \exp\left[-i\left(\omega_0'^2 - \gamma^2/4\right)^{1/2} t\right] \exp(-i\gamma/2 t) + x_p e^{-i\omega t} \quad (4.7)$$

For $\omega_0'^2 - \gamma^2/4 > 0$: defines the regime of weak damping.

The solution of the homogeneous equation describes a transient feature

The second part in Eq (4.7) is a forced oscillation, substitute it into (4.6):

$$x_p = \frac{eE_0}{m} (\omega_0'^2 - \omega^2 - i\omega\gamma)^{-1} \quad (4.9)$$

The dipole moment:

$$p_x = ex_p \quad (4.10)$$

- The polarizability: $\hat{\alpha}(\omega) = \frac{ex_p}{E_0} = \frac{e^2}{m} (\omega_0'^2 - \omega^2 - i\omega\gamma)^{-1}$

- The polarization density $\mathbf{P}$ with amplitude $\mathbf{P}_0$:

$$\mathbf{P}_0 = N\hat{\alpha}\mathbf{E}_0 = \frac{Ne^2}{m} (\omega_0'^2 - \omega^2 - i\omega\gamma)^{-1} \mathbf{E}_0 \quad (4.11)$$

- This means that we describe the ensemble of oscillators as an effective medium, an approach which is well justified for $a \ll \lambda$

- From (4.12) and (2.27): we get for the dielectric displacement:

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} = \epsilon_0 \left[1 + \frac{Ne^2}{\epsilon_0 m} (\omega_0'^2 - \omega^2 - i\omega\gamma)^{-1} \right] \mathbf{E} \quad (4.13)$$

$$\epsilon(\omega) = 1 + \left(\frac{Ne^2}{\epsilon_0 m} \right) (\omega_0'^2 - \omega^2 - i\omega\gamma)^{-1} = \chi(\omega) + 1 \quad (4.14)$$

介电常数 = 1 + 极化率

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- The spectral region between ω_0 and ω_L is often called "Reststrahlenbande" for the following reason. If we send a light beam with a broad, essentially flat, spectral distribution of the intensity around ω_0 onto a sample and allow it to be reflected several times, e.g., in the configuration given in Fig. 4.7a, then significant intensity will remain as a "rest" only in the region

$$\omega_0 \leq \omega \leq \omega_L$$

- It should be noted that we do not assume any coherent superposition of the beams, i.e., the Reststrahlenbande comes only from $\epsilon(\omega)$ and has nothing to do with Fabry–Perot resonators

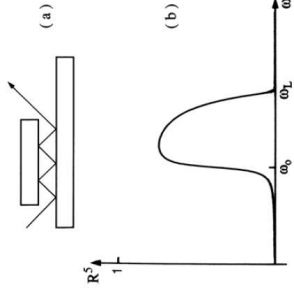


Fig. 4.7. An arrangement for multiple reflection from a medium (a) and the resulting "Reststrahlenbande" (b)

α_c

Semiconductor slab

- Handle semiconductor slabs with two surfaces and a geometrical thickness d
- (a) No interference
- (b) Interference

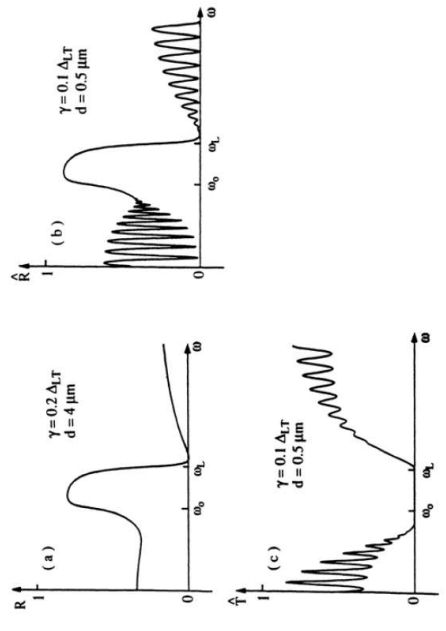


Fig. 4.8. The reflectivity R of a thin slab of matter in the vicinity of a resonance for the cases where intensities (a) or field amplitudes (b) have to be added and the transmission (b) for the case where amplitudes add